

# Sinapsistencia: A Web Platform for Medical-Legal Mediation with Explainable Machine Learning in the Peruvian Health Sector

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## Abstract:

Healthcare professionals in Peru face growing exposure to medical malpractice claims and often lack structured, timely access to specialized legal advice. Informal channels delay case management and reduce traceability. This study presents Sinapsistencia, a web platform that connects physicians with medical-law attorneys through a microservices architecture (Angular 21, Spring Boot 3.5, PostgreSQL, and a FastAPI machine learning service). The system supports medical-legal case registration with simulated clinical context compliant with Peruvian data protection law (Law 29733), document traceability, role-based dashboards, and two explainable ML components: a Random Forest model for legal risk assessment and a TF-IDF cosine-similarity matcher for attorney recommendation. The risk model was trained on 6,000 synthetic cases using a rule-based label generator and achieved macro precision of 0.618, recall of 0.597, and F1-score of 0.603 on a held-out validation set. Functional validation included five automated integration test suites and incremental execution of acceptance scenarios. Average inference latency for risk assessment remained below 200 ms in local experiments. Results indicate that a hybrid ML approach with explicit XAI factors and cold-start fallbacks is feasible for academic MVP validation in a sensitive domain where AI must act as decision support rather than autonomous judgment.

## 1 INTRODUCTION

Medical malpractice and negligence disputes in Peru are increasingly visible through institutional supervision and patient advocacy channels. SUSALUD reported more than 163,000 health-rights attentions in 2024 alone, reflecting sustained demand for formal recourse mechanisms. Physicians, however, frequently rely on informal networks to obtain legal orientation, with estimated response delays of 48 to 72 hours and limited audit trails (Miziara & Miziara, 2022; Tumelty, 2023).

This problem is relevant because delayed legal guidance increases professional vulnerability, encourages defensive medicine, and hampers early case structuring (Gunenc et al., 2024; Rowland et al., 2022). Digital platforms could improve mediation, yet naive approaches fail for three reasons: (i) medical-legal data are sensitive and regulated; (ii) black-box recommendations are ethically unacceptable in liability contexts (Jones et al., 2023; Scott et al., 2024); and (iii) cold-start conditions limit collaborative filtering when historical attorney-case data are scarce.

Prior solutions address fragments of the problem—electronic records, generic legal marketplaces, or clinical decision support—but not integrated physician-to-attorney mediation with explainable prioritization under role-based governance. Sinapsistencia differs by combining (1) simulated non-identifiable case context, (2) microservice separation of ML inference, (3) persisted XAI explanations, and (4) specialty-based fallback matching when the ML service is offline.

The key components of our approach are: a three-portal web application (physician, attorney, administrator); a Spring Boot API with ownership enforcement replacing database-level RLS; a FastAPI service implementing Random Forest risk scoring and TF-IDF attorney ranking; and an audit trail for critical actions. Current limitations include synthetic training data, pilot validation without real patient records, and pending large-scale functional test execution against a 90-case academic test plan.

The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 describes the proposed architecture and algorithms; Section 4 explains the validation methodology; Section 5 presents results; and Section 6 concludes with future work.

## 2 RELATED WORKS

Research on medical-legal risk, AI-assisted decision support, and recommender systems provides the foundation for Sinapsistencia. We group prior efforts into three lines.

### **Medical negligence and digital liability.**

Miziara and Miziara (2022) review medical errors, negligence, and defensive medicine, highlighting documentation and communication as recurring failure points. Rowland et al. (2022) analyze digital health-specific malpractice risks, emphasizing traceability and accountability. Wong et al. (2021) examine emergency department claims, showing that procedural and documentation issues dominate litigation patterns. These studies justify structured case intake and document versioning but do not provide integrated mediation workflows between clinicians and attorneys.

### **Trustworthy AI in clinical and legal contexts.**

Jones et al. (2023) report clinicians' concerns about trust and liability when using AI decision support. Wang et al. (2023) systematically review human-centered design for AI-enabled clinical systems, recommending transparency and explicit limitations. Scott et al. (2024) further note adoption barriers when explanations are absent. Our platform addresses this gap by persisting risk factors derived from feature importance and by displaying non-deterministic advisory messages in the user interface.

### **Matching and risk modeling techniques.**

Manning et al. (2008) and Ricci et al. (2015) establish TF-IDF and content-based recommendation as effective cold-start strategies—relevant for attorney matching without dense interaction logs. Random Forest classifiers remain a robust baseline for tabular risk scoring with interpretable feature contributions (Aggarwal, 2016). Unlike prior legal-tech marketplaces, Sinapsistencia couples both techniques in a health-specific workflow with simulated data aligned to Peruvian privacy constraints (Congreso de la República del Perú, 2011).

## 3 PROPOSED SOLUTION

Sinapsistencia follows a decoupled microservices architecture. The Angular 21 frontend consumes REST APIs from a Spring Boot 3.5 backend (Java 21) secured with JWT and role-based access control. PostgreSQL 16 stores operational data and Flyway migrations version the schema. A dedicated FastAPI service hosts ML models, invoked by the backend through an HTTP proxy with a 5-second timeout and graceful degradation.

### **3.1 System Architecture**

The physician portal supports case creation with simulated legal-clinical context (reference age, medical area, summary, relevant factors) without identifiable patient data. The attorney portal lists relevant pending cases and contact requests. The administrator portal manages users, audit logs, ML metrics, and summary reports. Ownership rules ensure that physicians cannot access peers' cases—enforced in the service layer and validated by automated integration tests.

### **3.2 Risk Assessment Model (Random Forest)**

The risk module predicts `risk_score` (0–1) and `risk_level` (bajo, moderado, alto, critico) from structured features: medical specialty, procedure complexity, priority, documentation completeness, informed consent, prior complaints, and days since incident. Training data comprise 6,000 synthetic rows generated by business rules plus Gaussian noise to avoid trivial memorization. A `RandomForestClassifier` (200 trees) and `RandomForestRegressor` share the same preprocessed feature space (one-hot specialty, ordinal complexity/priority). Explainability combines global feature importance with normalized case-specific values to produce `risk_factors` shown to users as advisory context.

### **3.3 Attorney Matching (TF-IDF + Cosine Similarity)**

Attorney profiles are vectorized with `TfidfVectorizer` over legal specialties, medical areas of interest, and biographical text. Physician queries built from specialty and case context are compared via cosine similarity. Results include `content_score`, matched specialties, and natural-language reasons. When the ML service is unavailable, the backend applies a cold-start fallback ranking attorneys by medical-area overlap.

### **3.4 API Contract**

FastAPI exposes GET /health, GET /api/v1/model/info, POST /api/v1/risk-assessment, and POST /api/v1/recommendations. Spring Boot proxies these endpoints and normalizes responses to camelCase for the frontend. High/critical risk levels trigger optional fire-and-forget notifications to an external automation webhook.

## 4 EXPERIMENTS AND VALIDATION

Validation followed four tracks aligned with the project quality plan: (1) ML offline evaluation, (2) backend integration testing, (3) frontend build verification, and (4) incremental functional scenarios from the academic test catalog.

### 4.1 Experimental Protocol for ML

The synthetic dataset was split 80/20 with stratification on risk\_level. Metrics reported on the test partition. Matching was evaluated qualitatively on demo scenarios with a corpus of eight seeded attorney profiles aligned to database identifiers. All experiments ran on a development workstation with Python 3.12, scikit-learn 1.5.2, and FastAPI 0.115.0.

### 4.2 Software Integration Tests

Five JUnit suites executed against Testcontainers PostgreSQL: application context load, ownership enforcement, case sub-resources (timeline and legal responses), feature flows (password change, status transitions, admin reports), and document upload/download. Frontend production build (ng build) verified compile-time integrity of physician, attorney, and admin portals.

## 5 RESULTS

### 5.1 Quantitative Results

Table 1 summarizes Random Forest performance on the held-out test set (model version rf-v1).

| Metric                         | Value                 |
|--------------------------------|-----------------------|
| Precision (macro)              | 0.6179                |
| Recall (macro)                 | 0.5973                |
| F1-score (macro)               | 0.6034                |
| Dataset size                   | 6,000 synthetic cases |
| Mean inference latency (local) | ~120 ms               |

Table 1: Risk classifier evaluation metrics.

All five backend integration suites passed (100% success) with Docker-enabled PostgreSQL. Critical REST operations—including login, case creation, matching proxy, and admin summary—completed under 2 seconds in local measurements, meeting the OE3 response-time indicator for the MVP scope.

### 5.2 Qualitative Results

End-to-end demonstrations confirmed: (i) role-based navigation across three portals; (ii) case registration with simulated context; (iii) attorney recommendations with XAI reasons; (iv) persisted ML classifications retrievable from the API; and (v) visible AI advisory disclaimers. Attorneys received relevant unassigned cases filtered by declared medical areas. These observations support feasibility for academic pilot evaluation at Clínica SANNA "El Golf" using simulated scenarios only.

## 6 CONCLUSIONS

This paper presented Sinapsistencia, a medical-legal mediation platform that integrates structured workflows, auditability, and explainable machine learning for the Peruvian health context. The hybrid design—Random Forest for risk scoring and TF-IDF for attorney matching—addresses cold-start constraints while keeping inference latency suitable for interactive use.

Results on synthetic data demonstrate moderate classification performance ( $F1 \approx 0.60$ ) and successful software integration across frontend, backend, database, and ML service layers. The system explicitly treats AI outputs as non-deterministic support, aligning with ethical requirements for liability-sensitive domains.

Future work includes: (i) expanding the attorney corpus and real pilot feedback without identifiable patient data; (ii) quantitative matching evaluation toward a  $\geq 70\%$  top-3 relevance target; (iii) cloud deployment for demo environments;

(iv) execution of the full 90-case functional test plan; and (v) usability assessment using the System Usability Scale in TP2.

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